INCH-POUND

MIL-DTL-55302/149C
18 February 2005
SUPERSEDING
MIL-C-55302/149B
17 November 1988

DETAIL SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: PLUG WITH SOCKET CONTACTS, 1.516 INCHES LENGTH, 2 KEYING PINS, FOR PRINTED WIRING BOARDS (.100 INCH SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-55302.

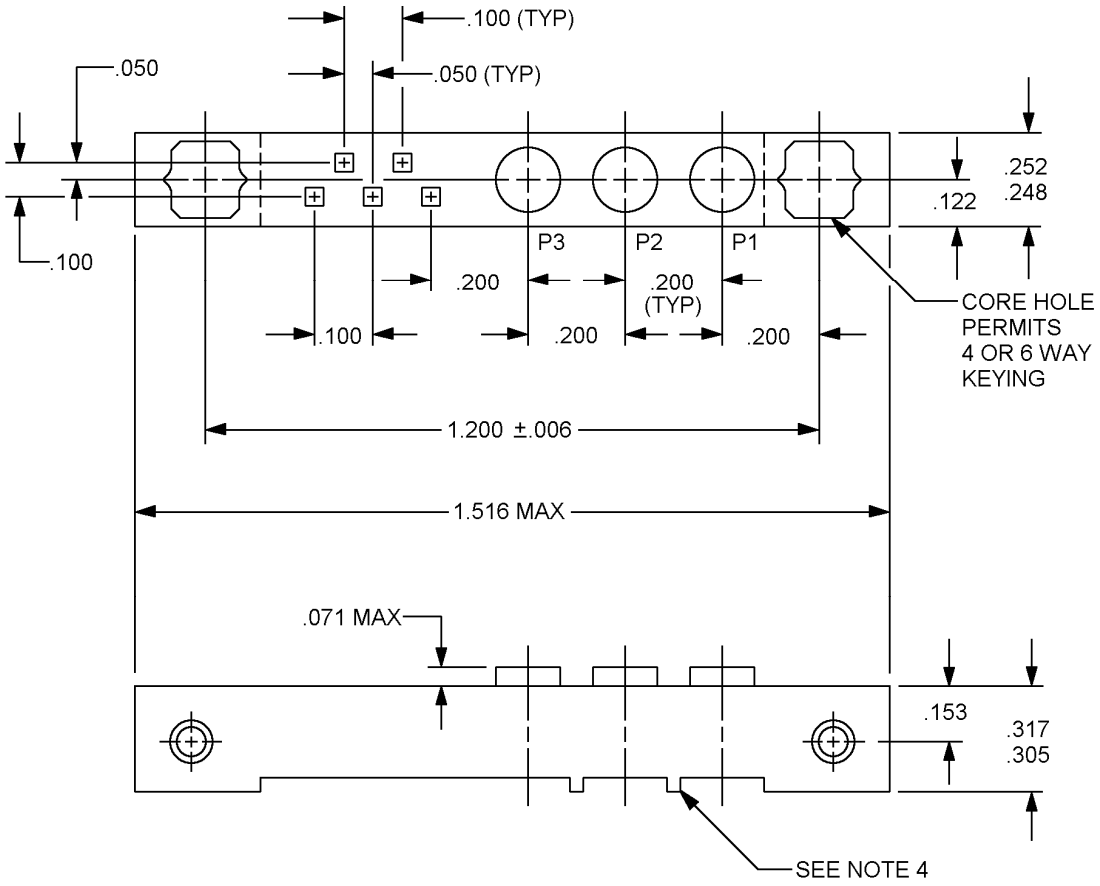


FIGURE 1. Connector, plug (.100 inch spacing).

AMSC N/A

FSC 5935

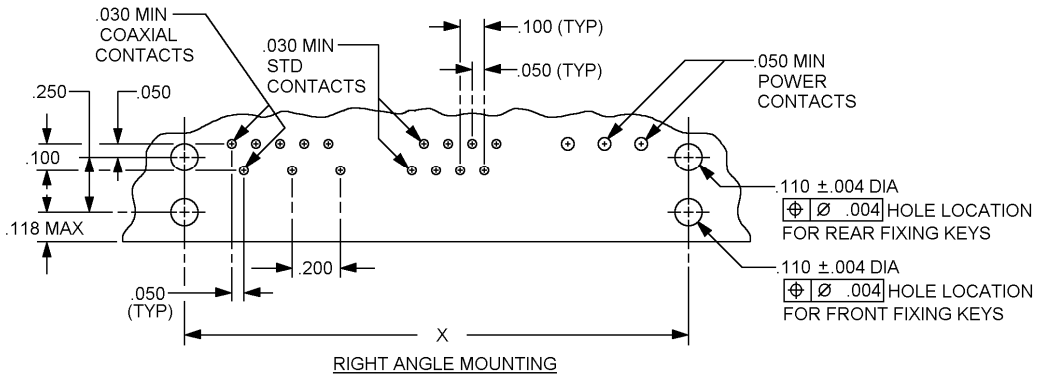
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Inches	mm	Inches	mm
.006	0.15	.248	6.30
.050	1.27	.252	6.40
.100	2.54	.305	7.75
.122	3.10	.317	8.52
.153	3.89	1.200	30.48
.200	5.08	1.516	38.51

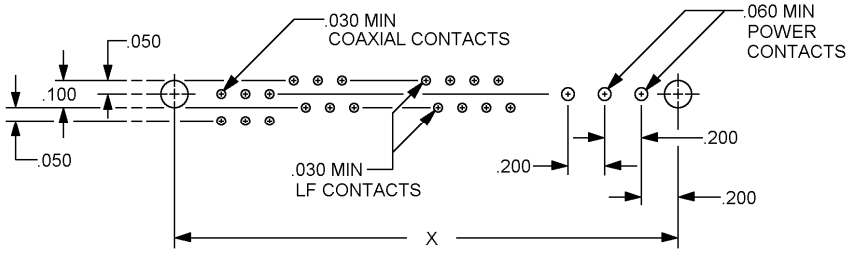
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ± .005 (0.13 mm).
4. Standoffs are located between each cable hole and between each hole and contact hole set.

FIGURE 1. Connector, plug (.100 inch spacing) - Continued.

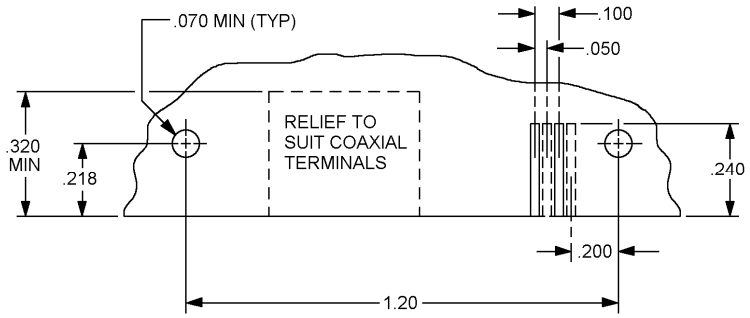
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RIGHT ANGLE MOUNTING



VERTICAL MOUNTING



EDGE CARD MOUNTING

Inches	mm	Inches	mm
.004	0.01	.110	2.79
.030	0.76	.118	3.00
.050	1.27	.120	3.05
.060	1.52	.200	5.08
.070	1.78.	.240	6.10
.100	2.54	.320	8.13
.106	2.69	1.20	30.5

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ± .005 (0.13 mm).

FIGURE 2. Recommended hole patterns.

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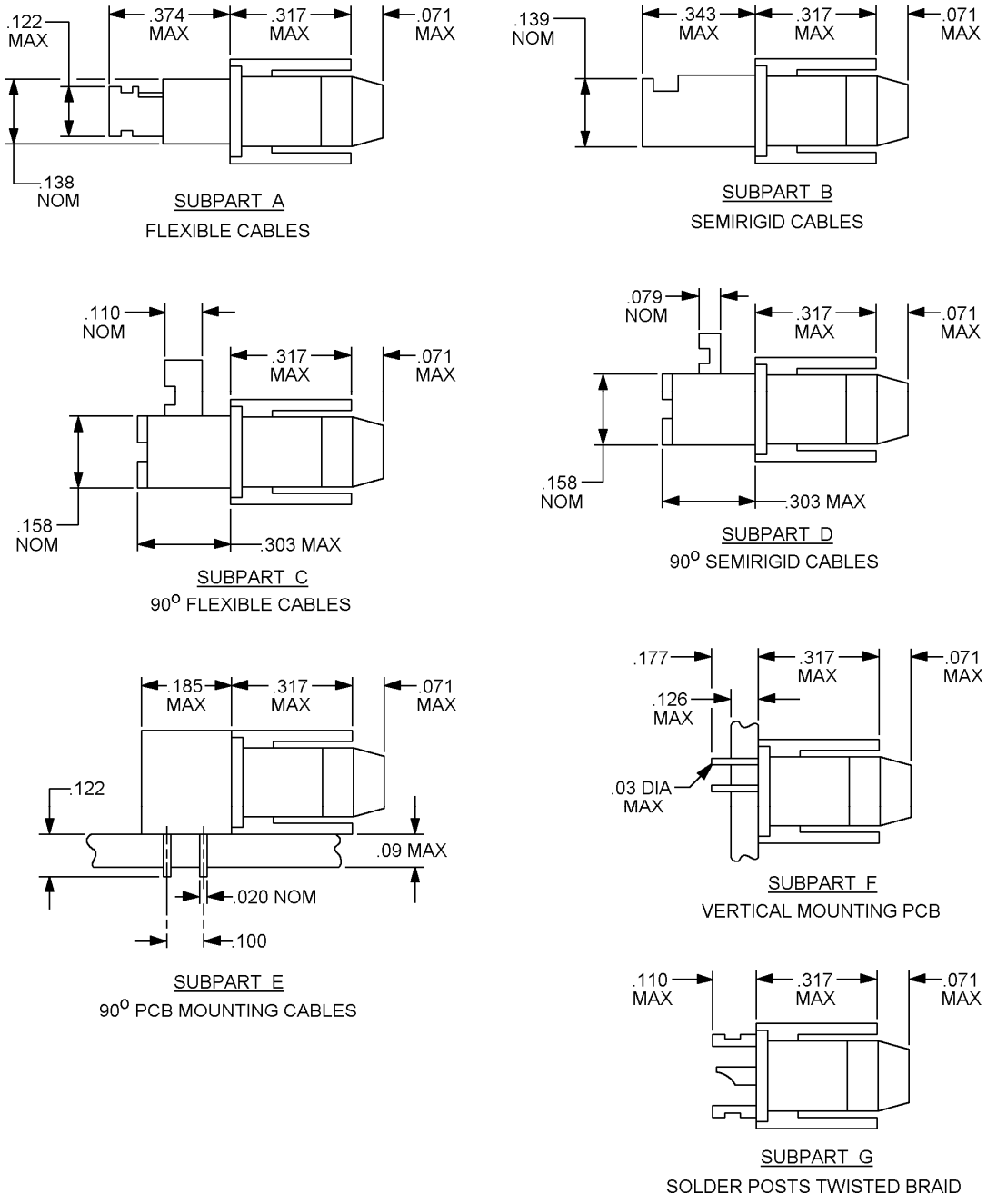
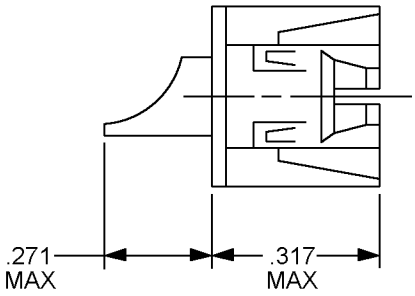
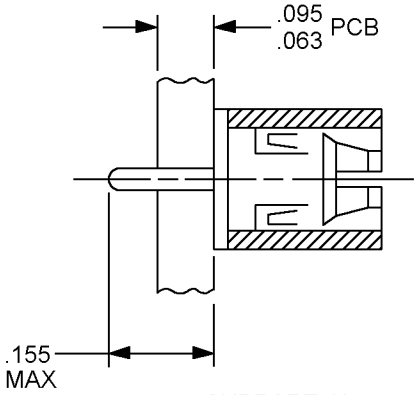


FIGURE 3. Coaxial configurations.

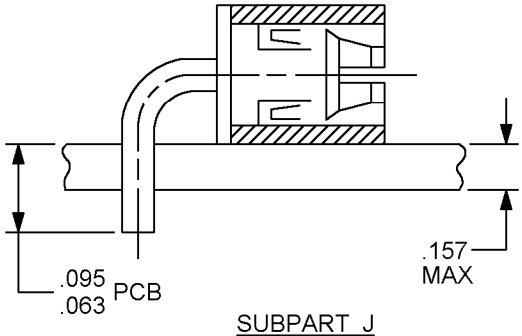
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SUBPART H
SOLDER BUCKET



SUBPART K
VERTICAL MOUNTING PCB



SUBPART J
90° PCB MOUNTING

Inches	mm	Inches	mm
.020	0.51	.138	3.51
.03	0.8	.139	3.53
.063	1.60	.155	3.94
.071	1.80	.157	3.99
.079	2.01	.158	4.01
.094	2.39	.177	4.50
.095	2.41	.185	4.70
.100	2.54	.271	6.88
.110	2.79	.303	7.70
.122	3.10	.317	8.05
.126	3.20	.343	8.71
		.374	9.50

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ± .005 (0.13 mm).

FIGURE 3. Coaxial configurations – Continued.

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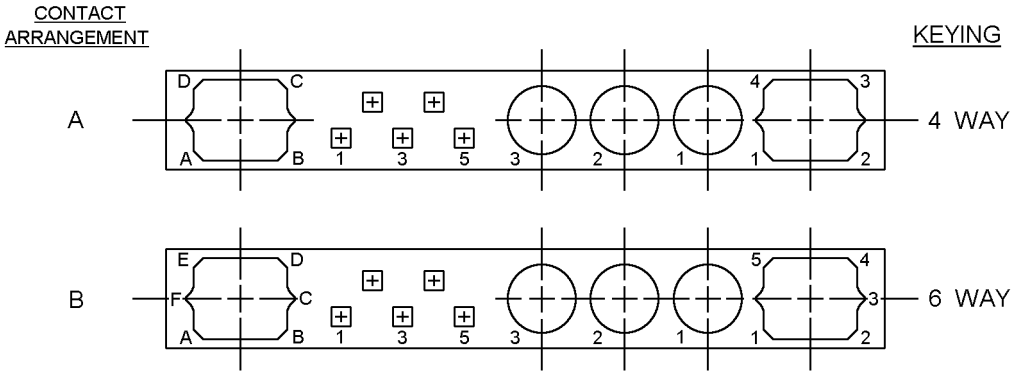


FIGURE 4. Contact and keying arrangements.

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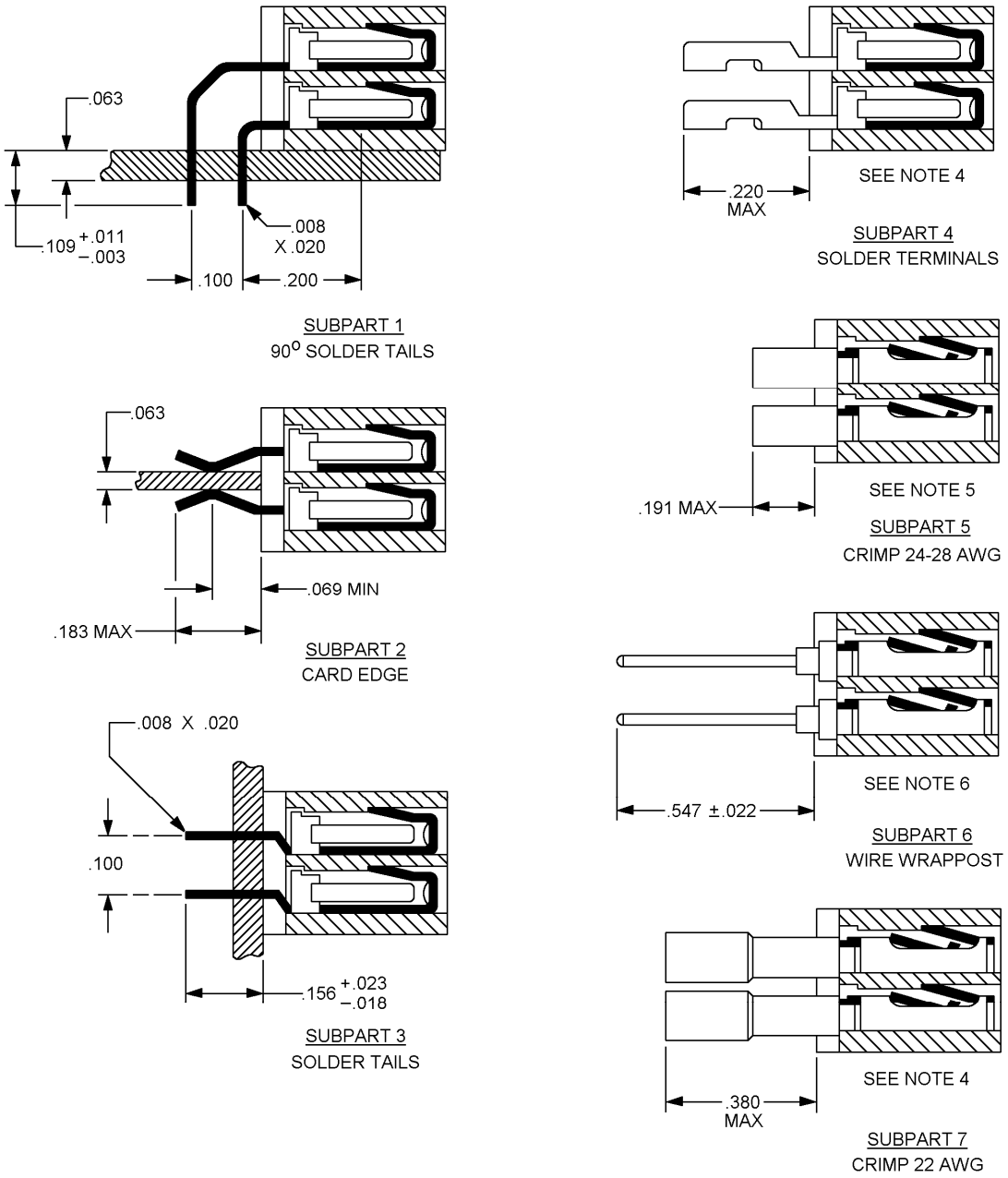


FIGURE 5. Contact termination.

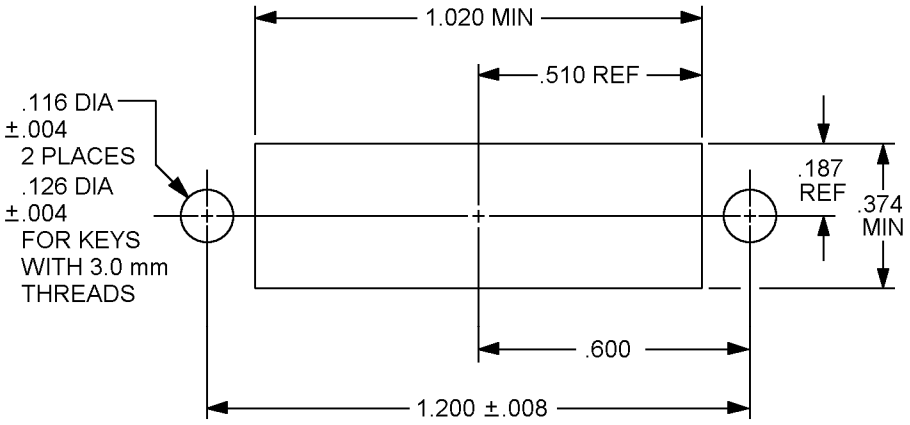
Source: https://assist.dla.mil -- Downloaded: 2026-10-22T22:52Z
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Inches	mm	Inches	mm
.003	0.08	.100	2.54
.008	0.20	.109	2.77
.011	0.28	.156	3.96
.018	0.46	.183	4.65
.020	0.51	.191	4.85
.022	0.56	.200	5.08
.023	0.58	.220	5.59
.063	1.60	.380	9.65
.069	1.75	.547	13.89

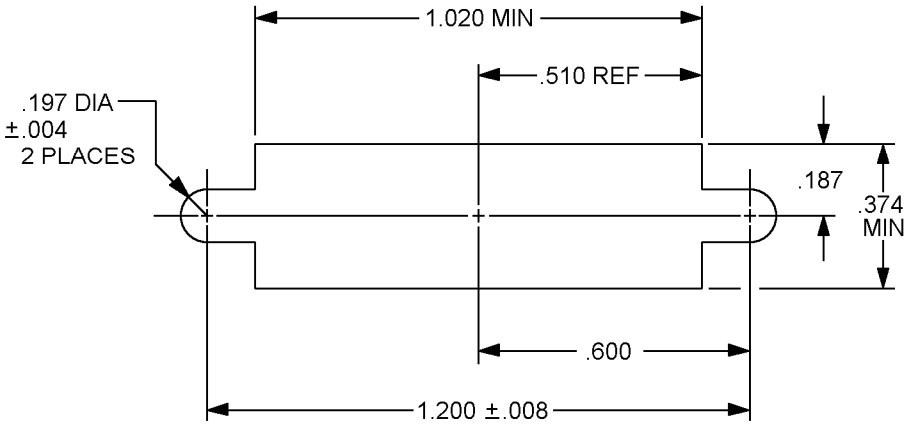
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ± .005 (0.13 mm).
4. Solder tail accepts no. 22 AWG wire.
5. Crimp contact accepts no. 22-26 AWG wire.
6. Contact with .025 $inch^2$ (0.64 mm^2) wrappost.

FIGURE 5. Contact termination - Continued.

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FIXED MOUNT



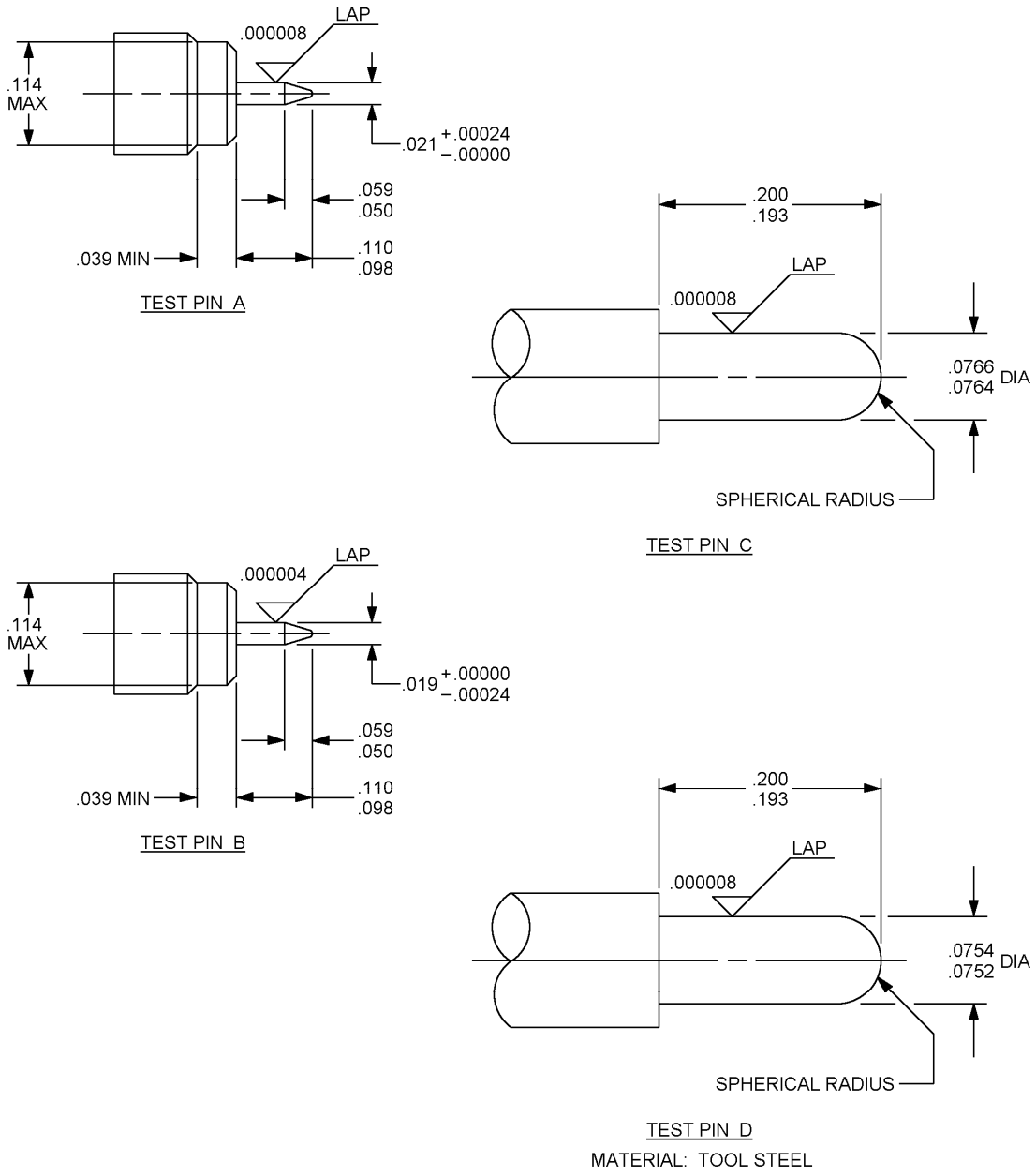
FLOATING MOUNT

Inches	mm	Inches	mm
.004	0.10	.374	9.50
.008	0.20	.510	12.95
.116	2.95	.600	15.24
.126	3.20	1.020	25.91
.187	4.75	1.200	30.48
.197	5.00		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ± .005 (0.13 mm).
4. Floating mount panel thickness .085 (2.15 mm) maximum.

FIGURE 6. Recommended panel cutouts.

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NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ± .005 (0.13 mm).

FIGURE 7. Test pins.

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REQUIREMENTS

Dimensions and configurations: See figures 1 through 7.

Contact identification: See figure 4.

Contact retention: Coaxial and power: 11.25 pounds minimum.
Signal: 4 pounds minimum.

Mating and unmating force (per contact):	Coaxial	Power	Signal
(Min)	7.2 ounces	3.6 ounces	0.7 ounce
(Max)	36 ounces	18 ounces	3.6 ounces

Values for each connector to be addition of contacts according to contact type.

Contact separation: 3.5 ounces minimum.

Contact resistance: 12 milliohms maximum.

Dielectric withstanding voltage:

Sea level: 1,000 V rms, 60 Hz.

High altitude: 300 V rms, 60 Hz.

Current rating:

Power contacts: 25 amperes at 25°C.

Coaxial contacts: .5 ampere at 25°C.

Signal contacts: 3 amperes at 25°C.

Keying: Moldings and mounting hardware provide 16 or 36 polarizing positions according to choice of 4 or 6 way capability. Keys are constructed in accordance with MIL-DTL-55302/154. Connectors are supplied with hex pins in the “1” and “A” positions, except for 1/4-turn locking styles, which are keyed in “F” and “6” positions. When required, connectors can be easily keyed to desired position by using a simple wrench.

Contact Separation

Signal 0.5 ounce minimum, measured using the test pins defined in MIL-DTL-55302/142.

Coaxial 0.5 ounce minimum using the following test pins

A. Insert and withdraw test pin A (see figure 7).
B. Insert test pin B fully and measure the separation force as the pin is withdrawn (see figure 7).

Power 1.4 ounce minimum using the following test pins

A. Insert and withdraw test pin C (see figure 7).
B. Insert test pin D fully and measure the separation force as the pin is withdrawn (see figure 7).

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Hood types: As specified in MIL-DTL-55302/155.

Mounting hardware: As specified in MIL-DTL-55302/154.

Mating connectors: As specified in MIL-DTL-55302/144.

Part or Identifying Number (PIN): Consists of the letter M, the basic number of the specification sheet, and a dash number compiled as follows:

Example:

M55302/149 * * * 2 *

Specification sheet number
Receptacle with pin contacts

Contact arrangement
(see figure 4)

Contact type – 3 amperes
(see figure 4)
O – Not applicable (see figure 4)

Contact configuration – coaxial/power (see figure 3)
X – less coaxial and power contact

Indexing type
2 – square post (16 positions)
1 – hex post (36 positions)

Mounting hardware
(see MIL-DTL-55302/154)

Previous note: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-55302, this document references the following:

MIL-DTL-55302/142 MIL-DTL-55302/154
MIL-DTL-55302/144 MIL-DTL-55302/155

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CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – EC
Air Force – 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-4494-000)

Review activities:
Army – AR, AT, AV, CR4, MI
Navy – AS, MC, OS, SH
Air Force – 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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